

COMPARISON OF PALM OIL FRUIT RIPENESS DETECTION AND CLASSIFICATION METHODS USING DEEP LEARNING

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Received October 2024; accepted December 2024

ABSTRACT. *Indonesia lacks skilled workers capable of determining the ripeness of oil palm fruits, which can affect oil palm production. In this evolving era, Artificial Intelligence (AI) has significantly improved ripeness detection systems, fostering innovation and competitiveness in agriculture. This paper presents an analysis and comparison of methods to determine Fresh Fruit Bunch (FFB) ripeness using YOLOv8s and YOLOv10s. Data were collected during a visit to an oil palm plantation operated by PTPN 4 in Medan, North Sumatra, Indonesia. 250 images were collected and manually classified into folders: “ripe”, “rotten” and “unripe”. Both models successfully identified the ripeness level of palm fruits, with YOLOv8 exhibiting better bounding box placement and standard confidence scores. YOLOv8’s effectiveness reached an mAP50 value of 0.995 after 25 and 50 training periods. Similarly, YOLOv10s reached values of 0.985 and 0.977 after 50 and 25 training periods. However overall, YOLOv8s has achieved better results in comparison for its accuracy.*

Keywords: YOLO, Ripeness detection, Deep learning, Oil palm fruit, Artificial intelligence

1. Introduction. Oil palm is the primary source of vegetable oil worldwide, due to its high productivity compared to other oil-producing plants. With over 50% of the world’s palm oil production coming from Indonesia [1], it plays a crucial role in both the Indonesian and global economies. In 2021, export volume increased by 0.6% with export value reaching US\$35 billion – 52% higher than the export value in 2020 of US\$22.9 billion [2]. Over a two-year period, industrial plantations expanded by 116,000 hectares, marking a 54% increase over the previous year. Regionally, the rise in deforestation linked to palm oil expansion was the most prominent in Kalimantan and Papua, while Sumatra experienced a somewhat smaller increase [3]. In 2021, export volume increased by 0.6% with export value reached US\$35 billion or 52% higher than the export value in 2020 at US\$22.9 billion [2]. Industrial plantations expanded by 116,000 hectares over two years, representing a 54% increase over the year before. Upon regional analysis, the increase in deforestation due to palm oil expansion was most significant in Kalimantan and Papua, with a slightly lower rise observed in Sumatra [3].

However, despite the high productivity, oil palm encounters difficulties in its processing, specifically in determining the classification of ripeness. The fruits on a single oil palm tree or plantation tend to differ in their stages of ripeness, complicating the sorting process because only ripe fruits should be harvested to maximize oil quality and productivity. Additionally, sorting oil palm fruits requires a large labor force. Identifying and separating ripe fruits from unripe ones requires skill and precision, and in some circumstances, a lack

of competent workers can be a considerable impediment. In Indonesia, the number of workers with expertise or skilled at determining the ripeness of oil palm fruits is still limited, which can hinder palm oil productivity.

Artificial intelligence technologies are increasingly being used in the agricultural sector due to their popularity. The shift towards AI-driven solutions, especially in the palm oil sector, is expected to improve efficiency and precision while meeting the rising need for advanced agricultural practices. Recent innovations in artificial intelligence, as mentioned in [4], have contributed significantly to improving the accuracy and scalability of ripeness detection systems, thereby fostering innovation and competitiveness in the agricultural industry.

This paper also presents an analysis and comparison of the different approaches and methods for analyzing the ripeness of FFBs (Fresh Fruit Bunches) by the previous researchers. The methods explored by previous researchers include R-CNNs (Regional-based Convolutional Neural Networks) [5], CNNs (Convolutional Neural Networks) [4,6,7], KNN (K-Nearest Neighbor) [8,9], ResNet50 [9], YOLO (You Only Look Once) [9-14], EfficientDet4 [10], DenseNet [15,16,22], and AlexNet [16]. These algorithms provide significant advances in accuracy and efficiency. R-CNN and CNN models frequently require extensive computer resources and long training times [5]. With large data sets, KNN may be memory-consuming and inefficient for prediction [8]. ResNet50, while powerful, is prone to overfitting and requires careful tuning [4]. Although YOLO is fast, it may struggle with small objects and precise localization [9]. EfficientDet4 and DenseNet can be difficult to implement and optimize (with a slightly lower accuracy), which results in increasing costs [10]. AlexNet, as an older architecture, may not perform as well as newer versions in terms of accuracy and efficiency [6].

Due to YOLO's popularity as the algorithm for object detection, the purpose of this research is to compare the detection accuracy of palm oil fruit ripeness using YOLOv8s and a newly developed algorithm, YOLOv10s, which is based on YOLOv8s. The decision is based on previous research comparing YOLOv7 with YOLOv8, which claims that YOLOv8 has improved capabilities, especially in identifying small objects [17]. Although in the study of [18], which still uses YOLOv7 to detect objects in real time, it can still achieve quite high accuracy.

This paper is structured into several chapters. The introduction presents an overview of the core idea of this research. The literature review summarizes existing research on palm oil fruit ripeness detection, comparing various deep learning models, especially those using the YOLO algorithm. The methodology describes the datasets, research setup, and deep learning algorithms employed in this study, including data preprocessing and model training procedures. The results and discussion present the findings of the research, analyzing the performance of different models in terms of accuracy, precision, recall, and other relevant metrics. The results will then be discussed, suggesting directions for improvement. This paper concludes with a summary of key findings of the study and suggests directions for future research.

2. Literature Review. In the past few years, palm oil ripeness detection has been developed through the integration of various learning techniques, such as those presented in [19] relying on the K-Nearest Neighbor (KNN) algorithm and achieving an average accuracy of 65% on a limited dataset. Although these methods have great potential for machine learning, more advanced techniques are still required.

Convolutional Neural Networks (CNNs) have been frequently used in recent research. For instance, [4] applied CNNs to larger datasets and achieved a significant accuracy improvement of 76% compared to traditional methods. Another recent advance was documented in [6,7], which achieved accuracies of 95.6% and 98%, by using CNNs on datasets

of over 600 images. This research underscores the effectiveness of CNNs in handling complex image classification tasks.

In another study [5], Faster R-CNN achieved 80% accuracy with a smaller dataset of 100 images, highlighting its capability in object detection [9]. When compared to YOLOv3 and ResNet50 on a dataset of 299 images, Faster R-CNN achieved an accuracy of 80%, while YOLOv3 outperformed ResNet50, achieving 84.7% accuracy, compared to ResNet50's accuracy of 76%. This demonstrates that YOLOv3 is more effective in real-time object detection tasks. Comprehensive research by [10] compared YOLOv8 and EfficientDet4 using a large dataset comprising 8299 images. YOLOv8 has achieved an impressive accuracy of 97.6%, underscoring the efficiency and advancements of YOLO models, EfficientDet4 closely followed with 94%. The research by [16] compared DenseNet with AlexNet on a dataset of 400 images, achieving accuracies of 86% and 77%.

Furthermore, the research by [15] achieved 92.5% accuracy by using DenseNet on a high-resolution dataset of 160 images, categorized into raw, under-ripe, and ripe stages, sized at 3264×2448 pixels and stored in JPEG format. This further validated the effectiveness of DenseNet architectures in image classification tasks. YOLOv4 has been used in other research, such as those by [11] and [19], and has achieved accuracies of up to 99%, indicating YOLOv4's potential in real-time applications. Overall, this research represents significant progress in the use of deep learning algorithms for palm oil fruit ripeness detection, which has been studied on numerous occasions. The YOLO method achieved the highest accuracy and shows great potential for practical applications in improving palm oil production efficiency. For further details on YOLO, refer to Table 1.

TABLE 1. Previous work of the YOLO model oil palm ripeness detection and classification

No	Author-Year	Dataset	Algorithm-Accuracy
1	N. Khamis et al. (2022) [9]	299 images, size 757×568 pixels	YOLOv3: 84.7%, ResNet50: 76%
2	M. R. A. Priyadi (2023) [10]	8299 images, size 640×640 and 1024×1024	YOLOv8: 97.6%, EfficientDet4: 94%
3	M. Asrol et al. (2023) [11]	45 images, size $52 \times 52 \times 3$, $26 \times 26 \times 33$, and $13 \times 13 \times 33$	YOLOv4: 85-99%
4	E. Salim (2023) [19]	4160 images	YOLOv4: 97-99%
5	J. W. Lai et al. (2022) [12]	240 positive images and 250 negative images, size 1920×1080	YOLOv4: 87.9%, ANN: 93%
6	N. A. M. B. Selvam et al. (2021) [13]	1500 images, size 3024×4032	YOLOv8: 99.49%
7	T. S. Gunawan et al. (2023) [14]	7171 images, size 416×416	YOLOv2: 87.9%

3. Methodology. In order to gather and process the data from the visit to one of PTPN 4's oil palm plantations in Medan, North Sumatra, Indonesia, multiple methods were used in this study. To get the best outcomes, these actions were necessary. The procedure comprised gathering datasets, pre- and post-processing images, and training and testing algorithms. The 's' models of YOLOv8 and YOLOv10 algorithms were employed in this research. The phases of putting research into practice are shown in Figure 1.

The data collection process involved capturing images of harvested palm fruits using an iPhone 12 camera. There were two different kinds of photos that were captured: close-ups of individual fruits and group photos of mature fruit clusters by visiting the palm plantation harvesting site to acquire these photos. For the training data, every fruit and every cluster of ripe fruits were photographed from several perspectives.

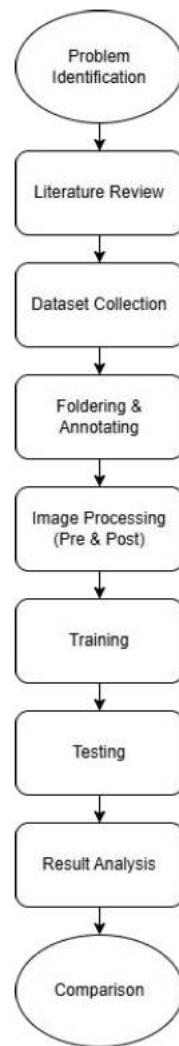


FIGURE 1. Research flowchart

During the dataset collection process, 250 images were collected and manually categorized into three folders: ripe, rotten, and unripe. While this categorization was not strictly necessary, it facilitated the upload and annotation process. The dataset was then uploaded to Roboflow, one folder at a time, with images annotated according to their pre-assigned categories. Out of the 250 images collected, we selected 80 of them based on their superior quality. These selected images underwent several augmentation post-processing steps, including flipping and rotating (15 and 90 degrees), resulting in a total of 185 images.

A popular real-time object recognition technique in computer vision and machine learning called YOLO (You Only Look Once) is used as the main algorithm for detection in this research, as it is known that YOLO is suitable for faster object detection and its accuracy [21]. YOLOv8s was built by Ultralytics while YOLOv10s was developed by the researchers at the Tsinghua University using YOLOv8s as the main base for the algorithm. YOLOv8s and YOLOv10s were the primary algorithms employed for comparison in this study. The previous researcher of YOLOv10s [20], claims that YOLOv10s was built using YOLOv8s as the base model, but with increased capability and efficiency. In comparison to the most recent YOLOv9-C version, it also has 25% fewer parameters (see Figure 2) [20].

4. Results and Discussion. To evaluate the performance of the algorithms, the tests were conducted by training and testing both YOLOv8s and YOLOv10s using the same

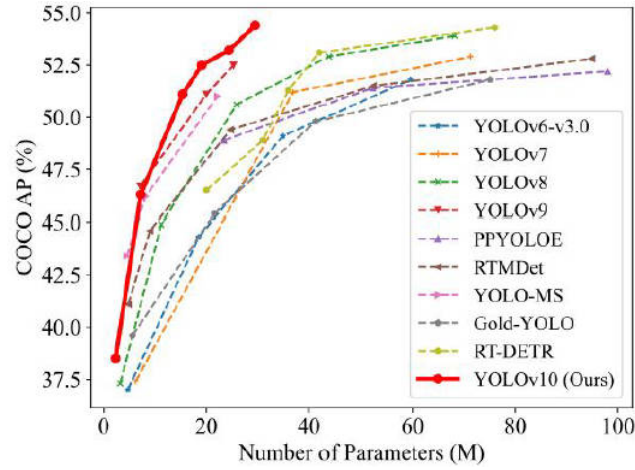


FIGURE 2. Number of YOLOv10 parameters versus others

TABLE 2. Comparison of training results YOLO

No	Algorithm	mAP50	mAP50-95	Epochs	GFLOPS
1	YOLOv8s	0.995	0.829	50	28.4
2	YOLOv8s	0.995	0.817	25	28.4
3	YOLOv10s	0.985	0.657	50	24.5
4	YOLOv10s	0.977	0.646	25	24.5

dataset. Each algorithm was trained for 25 and 50 epochs. Based on the training data results, we observed that YOLOv8s, trained for both 25 and 50 epochs, achieved the highest mAP50 scores of 0.995 out of 1. YOLOv10s, trained for 50 epochs, followed closely with a score of 0.985 out of 1. YOLOv8s also has a higher GFLOPS (Giga Floating Point Operations Per Second) indicating that it requires more computational resources for better accuracy compared to YOLOv10s that has lower GFLOPS. Lastly, YOLOv10s, trained for 25 epochs, obtained a score of 0.977 out of 1 (see Table 2 for reference).

From Figures 3 and 4, it clearly shows almost similar results in the metric graphs for both algorithms, with YOLOv8s yielding better results. For instance, in Recall, YOLOv8s performed better than YOLOv10s in correctly recognizing the relevant labels. It also outperformed YOLOv10s in precision, which measures the model's accuracy in prediction, scoring a point of 1 from 1, which YOLOv10s has never reached after running for 25 epochs.

Overall, in detection results, there is not much difference between the results. It is also notable that both results can still be improved with more datasets and epochs. The image

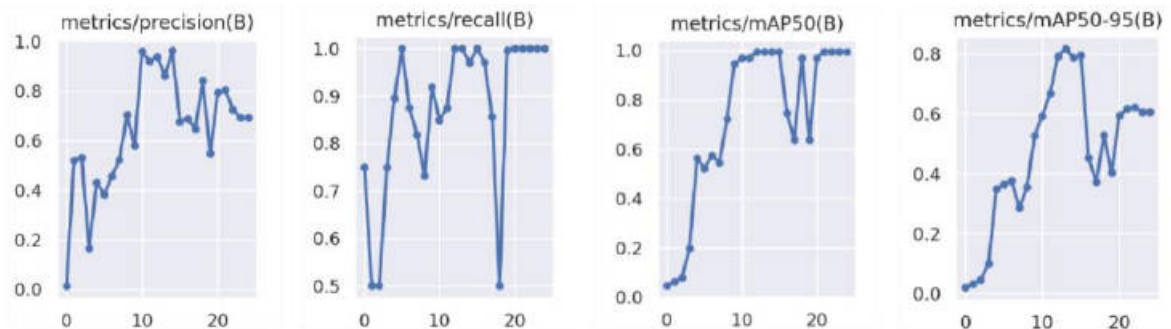


FIGURE 3. Metrics evaluation of YOLOv8s using 25 epochs

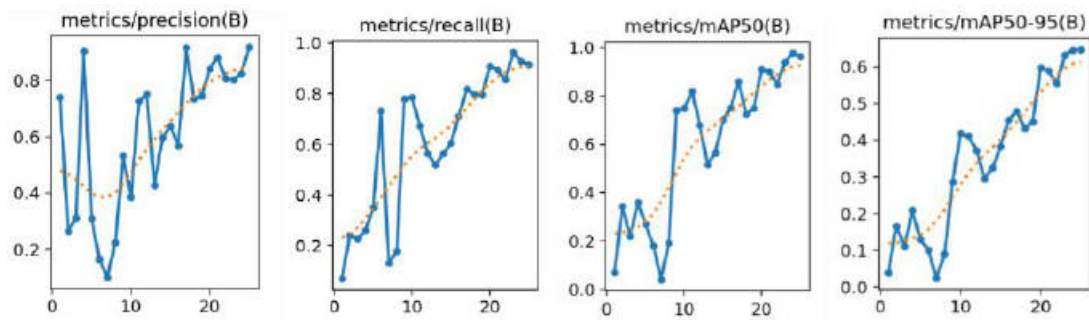


FIGURE 4. Metrics evaluation of YOLOv10s using 25 epochs



FIGURE 5. Ripeness detection using YOLOv8s



FIGURE 6. Ripeness detection using YOLOv10s

detection result using YOLOv8s and YOLOv10s algorithm can be seen in Figure 5 and Figure 6.

Compared to other results from the literature review process [4-16,19], it is proven that the results from this experiment that is YOLOv8s and YOLOv10s outperformed the other algorithms with each having 99% and 98% accuracy. In other results, YOLOv8s always performed consistently despite having different datasets and methods [2,6]. While YOLOv10s have a slightly lower 1% accuracy, it is notable that YOLOv10s also have a slightly lower GFLOPS (Giga Floating Point Operations Per Second) resulting in lower energy consumption using the same dataset and epochs.

5. Conclusions. This research focused on utilizing YOLOv8s and YOLOv10s algorithms to enhance the detection and classification accuracy of palm oil fruits. We gathered a dataset from an oil palm plantation in Medan, North Sumatra, Indonesia, consisting of 250 images. The images were categorized based on ripeness levels: ripe, rotten, and unripe. The images were augmented through techniques such as flipping and rotating to enhance the model's reliability.

YOLOv8s achieved an mAP50 score of 0.995 when trained for both 25 and 50 epochs. Similarly, YOLOv10s scored 0.985 after 50 epochs and 0.977 after 25 epochs. Notably, YOLOv10s exhibited greater efficiency with 24.5 GFLOPS compared to 28.4 GFLOPS for faster processing. Each detection results successfully identify ripeness classification of the palm fruit with a minimal difference, but YOLOv8s shows more accurate bounding box placements and higher confidence scores.

In conclusion, this research compared YOLOv8s with its improved version, YOLOv10s, to improve efficiency in oil palm production. Although the result showed no significant improvement in precision in ripeness detection, YOLOv10s require 25% fewer parameters. This research also promotes sustainable practices in palm oil production. Future studies could expand on these findings to investigate further enhancements and applications in agricultural fields.

Acknowledgment. The authors would like to express their appreciation to PTPN for supplying the essential data and allowing its use in this research.

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